



Cloud Computing

What is Cloud Computing?

Cloud computing is a computing paradigm in which tasks are assigned to a combination of connections, software and services accessed over a network. This network of servers and connections is collectively known as "the cloud."

Why is it so popular?

Computing at the scale of the cloud allows users to access supercomputer-level power. This vast processing power is made possible though distributed cluster computing, often in concert with virtualisation and parallel processing.

Who can use cloud computing?

Using a thin client or other access point, like an iPhone, BlackBerry or laptop, any users can reach into the cloud for resources as they need them.

How does it differ from Desktop computing?

Cloud computing allows users and companies to pay for and use only the services and storage that they need, when they need them.

Which organisations are implementing cloud computing?

Google has an initial edge in cloud computing precisely because of its need to produce instant, accurate results for millions of incoming search inquiries every day. Google Apps, Maps and Gmail are all based in the cloud.

Where is it heading?

New advances in processors, virtualization technology, disk storage, broadband Internet access and fast, inexpensive servers have all combined to make cloud computing a compelling paradigm. As a result, cloud computing has the potential to upend the software industry entirely, as applications are purchased, licensed and run over the network instead of your desktop.

Microsoft believe there is still space for innovation and are of the opinion that their telescopic pixel innovation would result in displays with much higher contrast ratios and brighter colours.

According to the scientists at Microsoft Labs, liquid crystals used in LCDs cannot completely block light in the off state and are hence unusable in bright light. LCDs transmit only 5 to 10 per cent of the backlight, because of the polariser, which blocks more than 50 per cent of the light.

Also, each colour filter transmits only 30 per cent of the remainder of the light, and there are some additional layers that decrease transmission even further. Telescopic pixels could also solve this problem.

These pixels are designed such that they have a pair of opposing mirrors, which completely contain light within a pixel and produce a perfect black colour. Using an electrical charge, the shape of one mirror can change its shape from planar to approximately parabolic, diverting the light beam to another mirror that will send the beam through the pixel and make the pixel visible on the outside. The researchers claim that this technology can transmit 36 per cent of the backlight, up from about 9 per cent in the most efficient technologies available today. The electrical charge could be applied very quickly, achieve display response times of 1.5 ms, the researchers said.

Considering the fact that there are already about 2 million pixels in today's HD displays and future displays may go more than

8 million (quad HD) or even 12 million (4K) pixels, installing mirrors in every pixels sounds complicated and expensive.

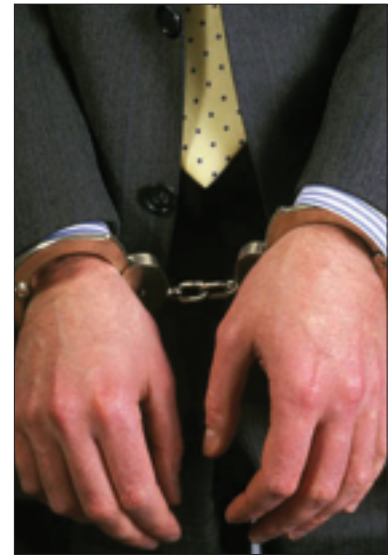
However, the scientists said that such a display could not only support a high image resolution, but can be made from relatively cheap materials, and is compatible with liquid crystal display production processes.

In addition to providing brighter computer displays, telescopic pixel technology could help drive forward the development of low-cost, large screen displays. The consumer market is demanding affordable, large, high-resolution televisions and displays, but this goal will be hard to achieve without using a modular approach, in which large displays are created by combining smaller building blocks.

PAY FOR YOUR SINS

Million Dollar Admin

A bail of \$5 million is not really an exaggeration, but exactly what a San Francisco admin recently got. Forty three year old Terry Childs is accused of taking illegal control of San Francisco from its FiberWAN network. The network apparently contains city e-mails, payroll, police records and information about prisoners—it was virtually an all-access pass to City Hall. Childs was arrested



because he refused to hand over passwords to the WAN system.

The FiberWAN network system is the major backbone of city government's computing infrastructure, connecting hundreds of different departments and buildings to a central data centre, and to each other. The FiberWAN system carries more than 60 per cent of the network traffic of all city government offices.

According to Ron Vinson, deputy director of the city's Department of Technology Information Services, Childs could have brought the whole network crashing down if he wanted to. However, this isn't the case as of now and the city's network continues to function normally. The reason that Childs' bail was set five times higher than most murder defendants' was because authorities were of the opinion that if released, he could permanently lock the system and erase records. One important fact is that Childs worked as a computer technician with the city for the five years.